

Program/Code:

```
1 // Import Node's readline module so the program can accept input from the console
2 const readline = require('readline');
3
4 // Create an input/output interface for reading user entries
5 const rl = readline.createInterface({
6   input: process.stdin,
7   output: process.stdout
8 });
9
10 // Store all valid integers entered by the user
11 let numbers = [];
12
13 console.log("Enter integers one at a time.");
14 console.log("Type q when finished.");
15
16 // Keep asking the user for input until they type q
17 function getInputs() {
18   rl.question("Enter an integer for q to quit: ", (input) => {
19     // Stop collecting input when the user enters q
20     if (input.toLowerCase() === 'q') {
21       // Handle the case where the user quits without entering numbers
22       if (numbers.length === 0) {
23         console.log("No numbers entered.");
24         rl.close();
25         return;
26       }
27       // Sort numbers from smallest to largest for the median calculation
28       numbers.sort((a, b) => a - b);
29
30       const count = numbers.length;
31
32       // Calculate the mean by adding all numbers and dividing by count
33       const sum = numbers.reduce((sum, num) => sum + num, 0) / count;
34
35       let median;
36
37       // Calculate median differently depending on odd or even count
38       if (count % 2 === 0) {
39         median = (numbers[count / 2 - 1] + numbers[count / 2]) / 2;
40       } else {
41         median = numbers[Math.floor(count / 2)];
42       }
43
44       // Find minimum and maximum values in the array
45       const min = Math.min(...numbers);
46       const max = Math.max(...numbers);
47
48       // Display final results
49       console.log("-----");
50       console.log("Count: ", count);
51       console.log("Mean: ", sum.toFixed(2));
52       console.log("Median: ", median);
53       console.log("Minimum: ", min);
54       console.log("Maximum: ", max);
55
56       rl.close();
57       return;
58     }
59
60     // Convert user input into a number
61     const number = Number(input);
62
63     // Reject invalid entries such as words or decimals
64     if (Number.isInteger(number)) {
65       console.log("Error: Please enter a valid integer.");
66     } else {
67       numbers.push(number);
68     }
69
70     // Continue asking for more input
71     getInputs();
72   });
73 }
74
75 getInputs();
```

Proof it runs successfully using node:

```
Last login: Tue Jun 23 19:08:03 on tty800
jacobbevans@new-host-5 ~ % cd ~/INFO465_Assignment1
node statistics.js
Enter integers one at a time.
Type q when finished.

[Enter an integer (or q to quit): 10
[Enter an integer (or q to quit): 20
[Enter an integer (or q to quit): 30
[Enter an integer (or q to quit): 40
[Enter an integer (or q to quit): q

Results
-----
Count: 4
Mean: 25.00
Median: 25
Minimum: 10
Maximum: 40
jacobbevans@new-host-5 INFO465_Assignment1 %
```

Proof code can handle errors:

```
Last login: Tue Jun 23 19:42:05 on ttys000
jacobevans@new-host-5 ~ % cd ~/INFO465_Assignment1
node statistics.js
Enter integers one at a time.
Type q when finished.

[Enter an integer (or q to quit): 10
[Enter an integer (or q to quit): hello
Error: Please enter a valid integer.
[Enter an integer (or q to quit): 20
[Enter an integer (or q to quit): 30
[Enter an integer (or q to quit): q

Results
-----
Count: 3
Mean: 20.00
Median: 20
Minimum: 10
Maximum: 30
jacobevans@new-host-5 INFO465_Assignment1 %
```

Local Repository Commit History (git log):

```
Last login: Tue Jun 23 18:13:11 on ttys000
jacobevans@new-host-5 ~ % git --version
git version 2.50.1 (Apple Git-155)
jacobevans@new-host-5 ~ % cd ~/INFO465_Assignment1
git init
git add statistics.js
git commit -m "Initial statistics program"
Initialized empty Git repository in /Users/jacobevans/INFO465_Assignment1/.git/
[main (root-commit) ad3c698] Initial statistics program
Committer: Jacob Evans <jacobevans@new-host-5.home>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

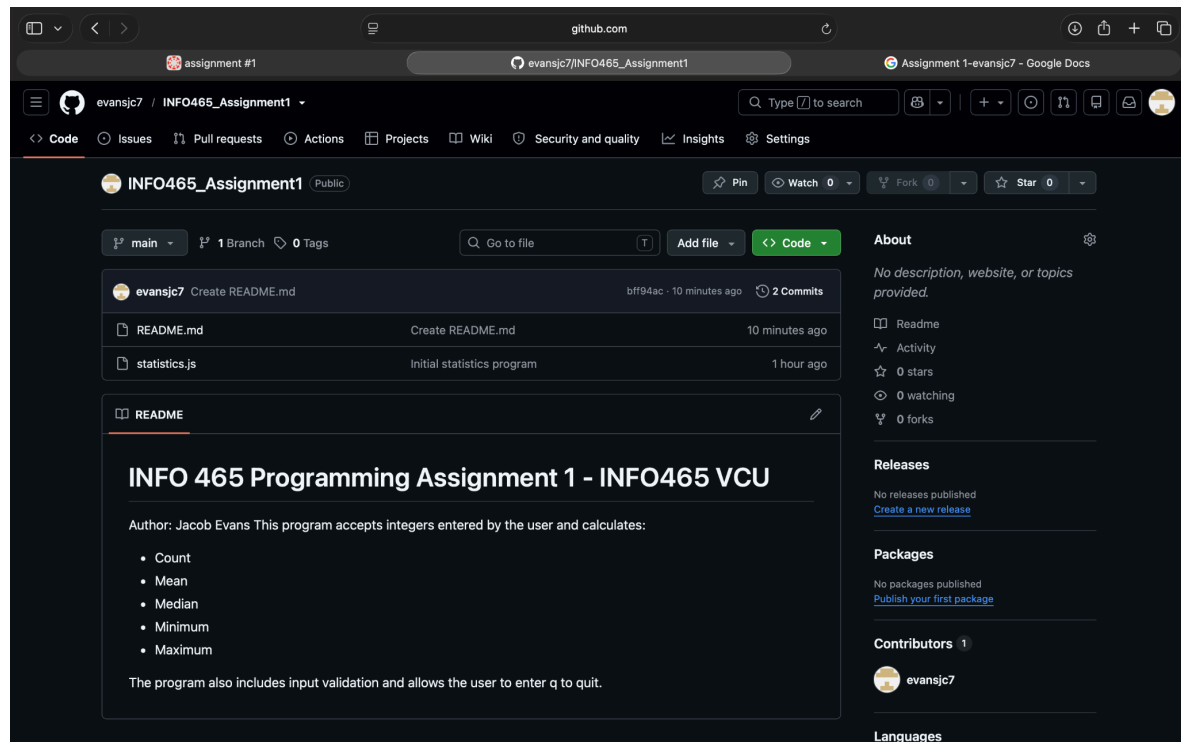
    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 79 insertions(+)
create mode 100644 statistics.js
jacobevans@new-host-5 INFO465_Assignment1 % git log --oneline
ad3c698 (HEAD -> main) Initial statistics program
jacobevans@new-host-5 INFO465_Assignment1 %
```

GitHub Repository with Program:



GitHub Repository with Collaborator:

